

Price Comparison as the First New Chatbot Action

DiscountMate - Deep Learning / AI Stream | Definition for Weeks 3-4

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Objective: Given a product, find equivalent listings across available retailers and identify the current cheapest price.

1. Action Scope

Price comparison will be the first new capability added while upgrading the existing Recipe RAG chatbot into a broader DiscountMate shopping assistant. The chatbot should recognise when a user wants to find or compare the price of a product, retrieve matching records from DiscountMate data, and return the comparison without inventing prices.

Supported user requests

- "Where is Coke Zero cheapest?"
- "Compare the price of Nutella."
- "How much is 2L Coke at Coles and Woolworths?"
- "Which store has the cheapest full cream milk?"
- "Is this cheaper at Coles or Woolworths?"
- "Show me the cheapest option."
- "What about the 1L version?"

2. Required Inputs / Entities

Entity	Required?	Example	Purpose
product_name	Yes*	Coke Zero	Primary search target
retailer	Optional	Coles	Restrict comparison to a retailer
brand	Optional	Coca-Cola	Improve product matching
pack_size	Optional	2L	Ensure comparable variants
category	Optional	Soft drinks	Disambiguation and filtering
product_id	Optional*	12345	Use current app/product context

* product_name is the minimum required input unless product_id is already supplied by the frontend/application context.

3. Expected Behaviour

1. Detect that the request is a price-comparison action.
2. Extract the available product information from the user message and application context.
3. Search DiscountMate for matching products.
4. Identify comparable variants of the requested product.
5. Retrieve current prices from available retailers.
6. Sort valid results from cheapest to most expensive.
7. Clearly identify the cheapest equivalent option.
8. Return structured data for the frontend.
9. Generate a short natural-language response using only retrieved product and price data.

Grounding rule: The LLM must never invent product prices. All prices shown to the user must come from DiscountMate data or an approved backend service.

4. Direct Comparison Rule

The chatbot should avoid presenting obviously different product variants as though they are identical. For Version 1, direct price comparisons should preferably match:

Product + Brand + Variant + Pack size			
Good direct comparison		Not a direct comparison	
Coke Zero 2L - Coles \$3.20 vs Woolworths \$3.50		Coke Zero 375mL can \$1.50 vs Coke Zero 2L bottle \$3.50	

5. Ambiguity Handling

If a request is too broad, the chatbot should not silently guess. For example, "Compare Coke" may match Coke Original, Coke Zero, Diet Coke and multiple pack sizes. The system should return candidate matches or ask a short clarification question such as:

"I found several Coke products. Are you looking for Coke Original or Coke Zero, and what size?"

More advanced clarification behaviour will be developed under DL-06-T15.

6. Proposed JSON Contract

```
{
  "action": "price_comparison",
  "query": {
    "product_name": "Coke Zero",
    "brand": "Coca-Cola",
    "pack_size": "2L",
    "retailers": []
  },
  "matched_product": {
    "product_id": "12345",
    "product_name": "Coca-Cola Coke Zero 2L",
    "brand": "Coca-Cola",
    "pack_size": "2L"
  },
  "prices": [
    {"retailer": "Coles", "price": 3.20, "currency": "AUD"},
    {"retailer": "Woolworths", "price": 3.50, "currency": "AUD"}
  ],
  "cheapest": {"retailer": "Coles", "price": 3.20},
  "status": "success"
}
```

7. Version 1 Boundaries

The first implementation should stay focused. The following features are intentionally outside the scope of DL-06-T2:

- Historical price trends
- Future price prediction
- Personalised recommendations
- Basket-level comparisons
- Product substitution
- Unit-price optimisation across different pack sizes
- Recipe-related price comparison

8. Updated Task Entry

Field	Definition
Reference	DL-06-T2
Sub-task	Define price comparison as the first new chatbot action, including supported queries, required entities, direct-comparison rules, ambiguity handling and structured JSON output.
Story Points	1
Period	Weeks 3-4
Dependencies	DL-06-T1
Risk	Product names, variants and pack sizes may be ambiguous, and some user requests may contain multiple actions or insufficient information for a direct comparison.